# Ana-Catalina Villalobos Contardo

## DATA SCIENTIST & MACHINE LEARNING ENGINEER

Metropolitan Region, Chile
anacatalina@outlook.cl • ana-catalina.com

## SUMMARY

Civil Engineer with a strong track record in Data Science, Machine Learning, Agentic AI, and Data Engineering. Combines a rigorous analytical and mathematical foundation with hands-on experience building agentic infrastructure powered by Model Context Protocol (MCP), high-availability distributed systems (geocoding, microservices), large-scale data pipelines (ETL/ELT), and ML solutions (predictive maintenance, operational optimization). Proven technical leadership managing multidisciplinary teams, focused on solving complex problems, operational reliability, and strategic impact.

## EXPERIENCE

**SimpliRoute** | *Learning Engineer* | August 2025 - Present

- Designed and deployed production agentic infrastructure using Model Context Protocol (MCP), featuring deterministic guardrails, least-privilege execution, and Redis/Kubernetes state persistence.
- Architected a high-availability, fault-tolerant multi-provider geolocation and geocoding platform using decoupled asynchronous microservices, thread-safe concurrency, and fallback strategies.
- Data modeling in Google BigQuery, distributed processing with Google Cloud Dataflow, and event orchestration with Google Pub/Sub and Apache Airflow, ensuring lineage and operational reliability.
- Automated retraining pipelines and operational monitoring for production Machine Learning predictive models.

**Fracttal** | *Tech Lead Fracttal Hub* | November 2023 - July 2025

- Technical leadership of an international, multidisciplinary integrations team and strategic roadmap planning for Fracttal Hub, aligning business objectives, mentoring, and continuous value delivery.
- Python development of data processing and task orchestration libraries featuring over 100 integration actions and advanced transformations across heterogeneous systems.
- Supervised and coordinated enterprise integration projects for dozens of clients, ensuring on-time delivery and high-reliability data pipelines.

**Fracttal** | *Data Scientist* | August 2021 - November 2023

- Designed and implemented data science processes for predictive maintenance, asset prioritization, and failure estimation.
- Collaborated on product creation and development to manage integrations between Fracttal and other platforms.

**Fracttal** | *Data Analyst* | February 2020 - July 2021

- Developed a predictive maintenance tool using advanced data analysis, predictive statistics, and Machine Learning techniques.
- Database design and management alongside web development tasks.

## TOOLS & TECHNICAL SKILLS

- **Programming Languages:** Python (Principal/Advanced) | SQL | TypeScript | JavaScript | MATLAB.
- **Cloud & DevOps:** Google Cloud Platform (BigQuery, Dataflow, Cloud Run, Artifact Registry, Google Pub/Sub, Cloud Storage, GKE) | ArgoCD | Git | GitHub | Bitbucket.
- **Frameworks & Libraries:** FastAPI | Pydantic | Pandas | NumPy | Scikit-learn | LangChain | TensorFlow | Django | React JS.
- **Databases:** PostgreSQL | Redis | BigQuery | SQL Server | MySQL | MongoDB | SQLite.
- **Containers & Orchestration:** Airflow | Airbyte | Docker | Kubernetes.
- **Methodologies & Management:** System Design | Test-Driven Development (TDD) | Agile Methodologies (Scrum & Kanban) | Roadmap Planning | Jira | Notion.
- **Key Skills:** Agentic AI & Multi-Agent Systems (MCP, LLMs) | Distributed Systems & Geocoding | Data Pipeline Architecture (ETL/ELT) | ML Model Monitoring | Relational DB Design | Technical Leadership.

## EDUCATION

- **Universidad de Chile** | *Civil Engineering with a minor in structures, construction, and geotechnics* | Graduated 2019, Degree awarded 2020 (Maximum Distinction)
  Thesis: Gaussian Processes and vibration modes for delamination detection in composite material plates.
- **Universidad de Chile** | *Bachelor of Engineering Sciences, Civil minor* | Completed 2017
  Notable activity: Teaching Assistant for Probabilities and Statistics.

## PUBLICATIONS

- **Generalized Gaussian smoothing for baseline-free debonding assessment of sandwich panels** | *Structural Control Health Monitoring* - March 2021 (https://doi.org/10.1002/stc.2727)
  Methodological proposal for delamination detection and evaluation in composite plates using Gaussian Processes and Bayesian analysis.

## LANGUAGES & CERTIFICATIONS

- **Languages:** Spanish (Native) | English (Advanced - EFSET English Certificate C2 Proficient, 2020)
- **Certifications:** React & Django Full Stack (Udemy, 2022) | Mastering Data Visualization (Udemy, 2022) | Applied Machine Learning in Python (Coursera, 2020).